

Quality control with control charts

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I would like to express my gratitude to Mr. Daniele Del Morino and Paola Del Morino for welcoming me as an intern in their company and providing invaluable support and guidance throughout my internship.

Company profile

The Del Morino company started Around 1875, with Domenico Del Morino commencing his journey as a blacksmith. For over 150 years, the family's unwavering commitment and dedication have transformed their artisanal venture into a company. Today the legacy is carried on by Daniele and Andrea Del Morino, who have embraced a business ethos centered around QUALITY and INNOVATION.

The factory spans across an expansive 10,000 m2 area, dedicated to the production of a diverse array of farm utensils, which encompass rotary tillers, lawn mowers, backhoes, and other equipment. The facility boasts a variety of cutting-edge machines, such as 3D laser cutting machines, automatic bending machines with CNC control, as well as robotic welding processes and automated electrostatic powder coating systems. These advanced technologies enable the factory to maintain high levels of precision and efficiency in its manufacturing processes.



Del Morino serves a global customer base, with a significant portion of their machines being sold in France. Additionally, their products find their way to various countries around the world, including Russia and Italy, catering to the diverse needs of international markets.

Introduction

Quality control is an important aspect for every production company, as it ensures the products manufactured to meet the expectations of customers. It includes a set of systematic processes that are designed to monitor, control, and enhance the quality of the product. By implementing effective quality control measures, companies can consistently deliver products that are reliable, safe, and meet the desired standards.

One of the statistical tools widely used in quality control is the control chart. Control charts are employed to monitor and analyse process variation over time. They offer visual representations of how a process performs, enabling teams to identify trends, patterns, and anomalies. These charts are instrumental in distinguishing whether a process is operating within acceptable limits (in control) or experiencing unexpected variations (out of control).

The application of control charts is versatile, making them valuable in numerous production areas. They can be effectively utilized in the assembly process, where they help track and maintain the accuracy and consistency of assembled components, leading to high-quality finished products. On welding machines, control charts aid in monitoring key parameters like weld strength and penetration, ensuring the welding process remains within specified parameters.

In the packaging sector, control charts can be employed to monitor critical factors such as weight, seal integrity, and label placement, ensuring that every packaged product meets the required quality standards before reaching customers. Additionally, control charts are equally applicable in the realm of material handling equipment, where they help track throughput, cycle times, and error rates, leading to better production optimization.

Furthermore, control charts are valuable in monitoring and improving the performance of other devices like injection moulding machines, where they help maintain consistent product dimensions and minimize defects during the manufacturing process.

In conclusion, control charts serve as powerful tools within the realm of quality control, offering insights into process performance and aiding in the continuous improvement of production processes across various areas in the company. By effectively implementing quality control measures and utilizing control charts, companies can enhance customer satisfaction, optimize manufacturing processes, and maintain a competitive edge in the market.

Possible applications

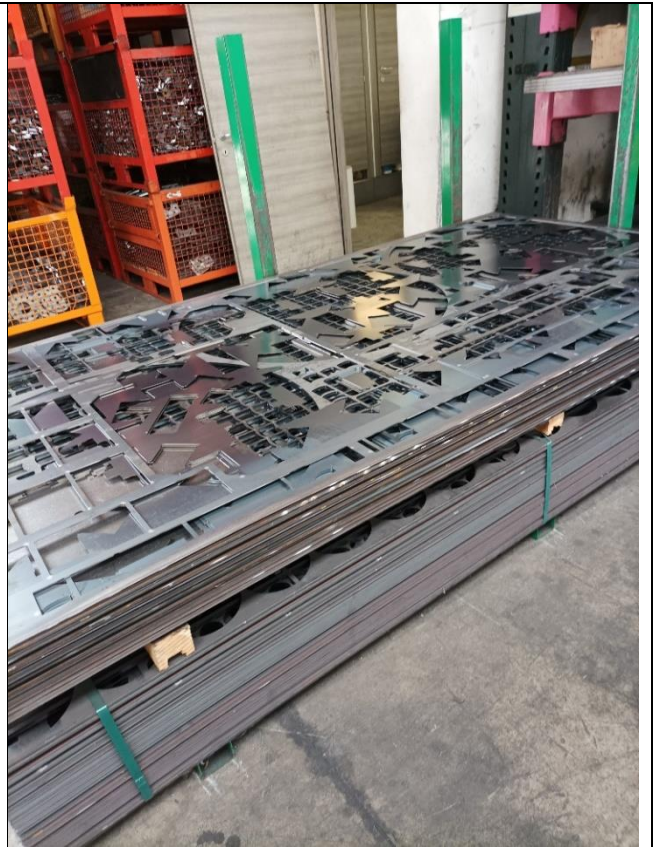
In our factory, the application of the control chart can be seen in various positions. First, in the moulding machines where there is a need of frequent replacement of the blade used to shape products, the control chart is easily applied to track the time till which the blade still meets the required precision to produce valid products.



This control chart can also be applied to the assembly and packaging sector to track any variation or defect of the final product and track packaging parameters to provide high quality products to every single customer.



Another place to apply the control chart in our factory would be in the material handling equipment, so the plaque cutting and bending devices, tracking their throughput, cycle time, or error rates allowing optimisation of the production process.



Project

During my internship in Del Morino's factory, the objective of my project was to apply a control chart, one of the different tools in quality control, on one of the production sectors of the factory. I chose to measure the pieces that forms the two ends of a rotor shaft that needs to be very precise to function perfectly. I chose to measure 30 pieces of URC172 and 40 pieces of URC176, they represent 1% of all of the pieces of the same kind in the stocks.



After all the measurements are done, the data are analysed.

Item: **URC 172**

measure	a (mm)	b (mm)	c (mm)
1	35.06	143.16	10.1
2	35	143.09	10
3	34.97	142.98	10.1
4	35.01	143.03	10
5	35.03	143.01	10
6	35.07	143.06	10
7	35.02	143.02	10.1
8	35	143.05	10.1
9	35	143.03	10
10	35.03	143.18	10
11	34.98	143.2	10.2
12	35.05	143.15	10
13	35.03	143.03	10
14	35	143.02	10
15	35.01	143.01	10.1
16	35.03	143	10
17	35.02	143	10
18	35.05	143.11	10
19	35	143.09	10.1
20	35.02	143.1	10
21	34.98	142.99	10
22	35.06	143.2	10.2
23	35.03	143.13	10
24	35.01	143.07	10
25	35.05	143	10
26	35.03	143	10
27	34.97	143	10
28	34.98	143	10.1
29	35.08	143.08	10
30	35	143.02	10

	Average	Standard Deviation
a	35.019	0.029402
b	143.0603	0.066565
c	10.03667	0.061495

Then, using the formula:

- Upper Limit (UL) = Average + 3*Standard Deviation
- Lower limit (LL) = Average - 3*Standard Deviation

We can get our upper limit and lower limit to construct the control chart.

	a	b	c
UL	35.10721	143.26	10.22115
LL	34.93079	142.8606	9.852182

Item: **URC 176**

measure	a	b	c
1	35.03	63.46	44.21
2	35.01	63.51	43.98
3	35	63.49	44.01
4	35.02	63.51	44.12
5	35.02	63.47	43.95
6	34.98	63.53	43.99
7	35	63.5	44
8	35	63.5	44.05
9	35	63.58	44.1
10	34.98	63.52	43.97
11	34.99	63.5	44.01
12	35.01	63.52	44
13	35.05	63.48	43.98
14	35.01	63.51	44
15	35.04	63.5	44.02
16	35.01	63.51	44.12
17	34.98	63.5	44.15
18	35.04	63.44	44.2
19	34.95	63.47	43.96
20	34.98	63.48	44
21	35	63.51	44.06
22	35.04	63.5	44
23	34.98	63.49	44.09
24	35	63.54	43.99
25	35.04	63.56	44
26	34.95	63.53	43.95
27	34.97	63.49	43.99
28	34.98	63.48	44.2
29	35.02	63.5	44.13
30	35	63.47	44.1
31	35.04	63.53	43.93
32	35	63.55	43.98
33	34.96	63.46	44
34	34.97	63.46	44.02
35	34.94	63.49	44.03
36	34.95	63.53	44
37	34.95	63.44	44.07
38	34.92	63.49	43.93
39	34.95	63.47	43.97
40	34.95	63.49	44

	Average	Standard Deviation
a	35.019	0.029402
b	143.0603	0.066565
c	10.03667	0.061495

Using the same formula:

- Upper Limit (UpL) = Average + 3*Standard Deviation
- Lower limit (LowL) = Average - 3*Standard Deviation

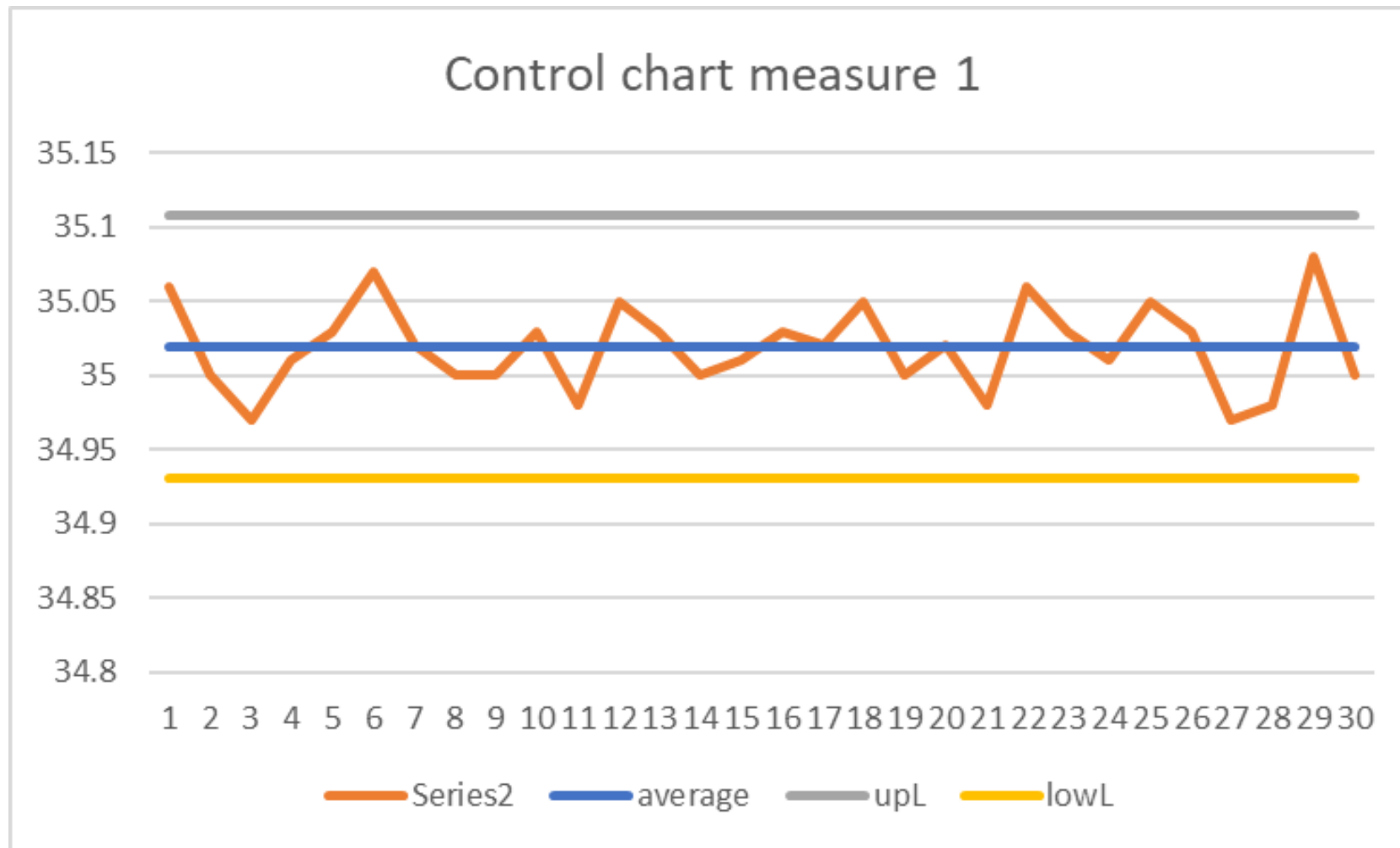
We can get our upper limit and lower limit to construct the control chart

	a	b	c
UL	35.09096	63.5906	44.25337
LL	34.89454	63.4074	43.80963

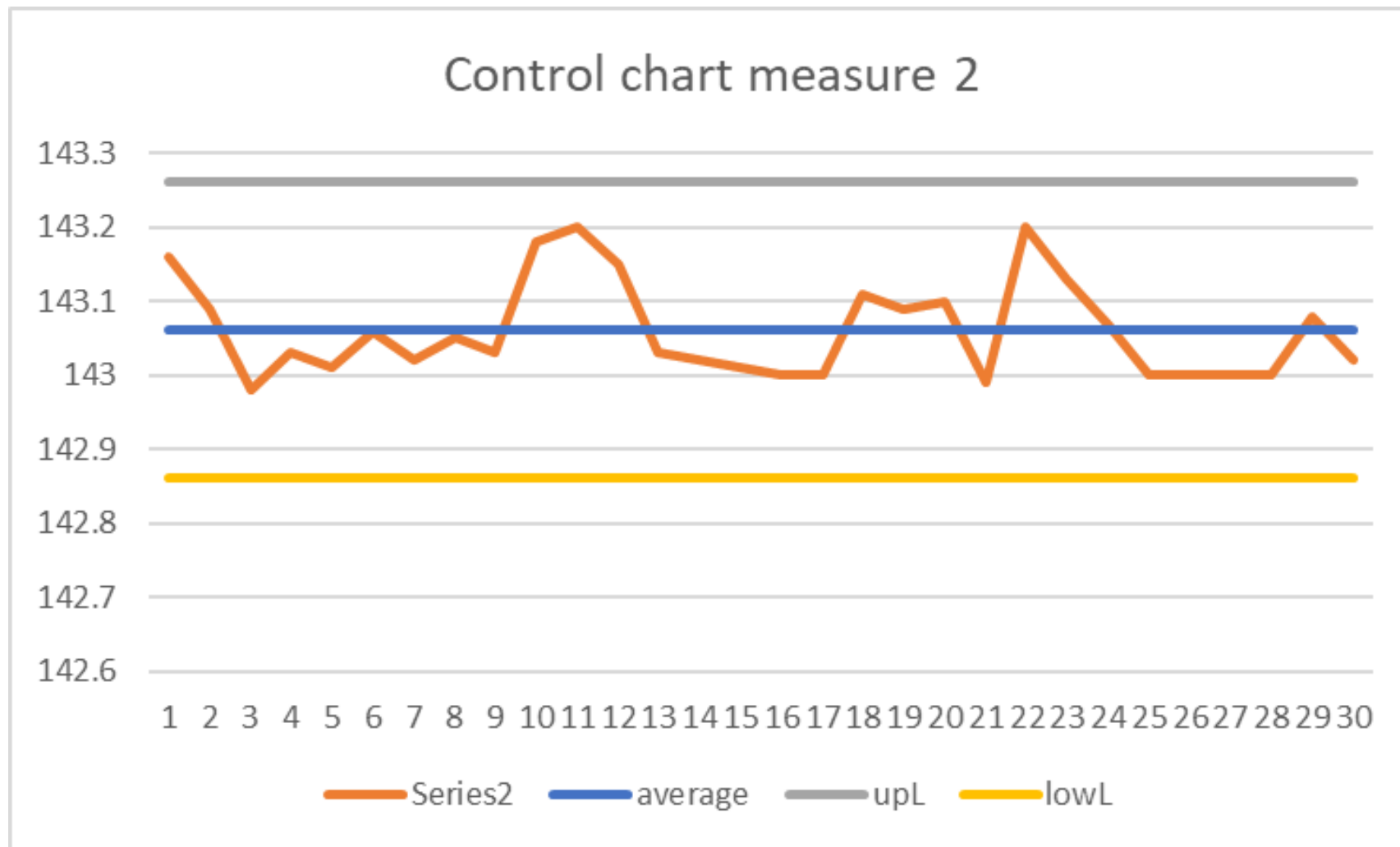
Control chart results

URC172

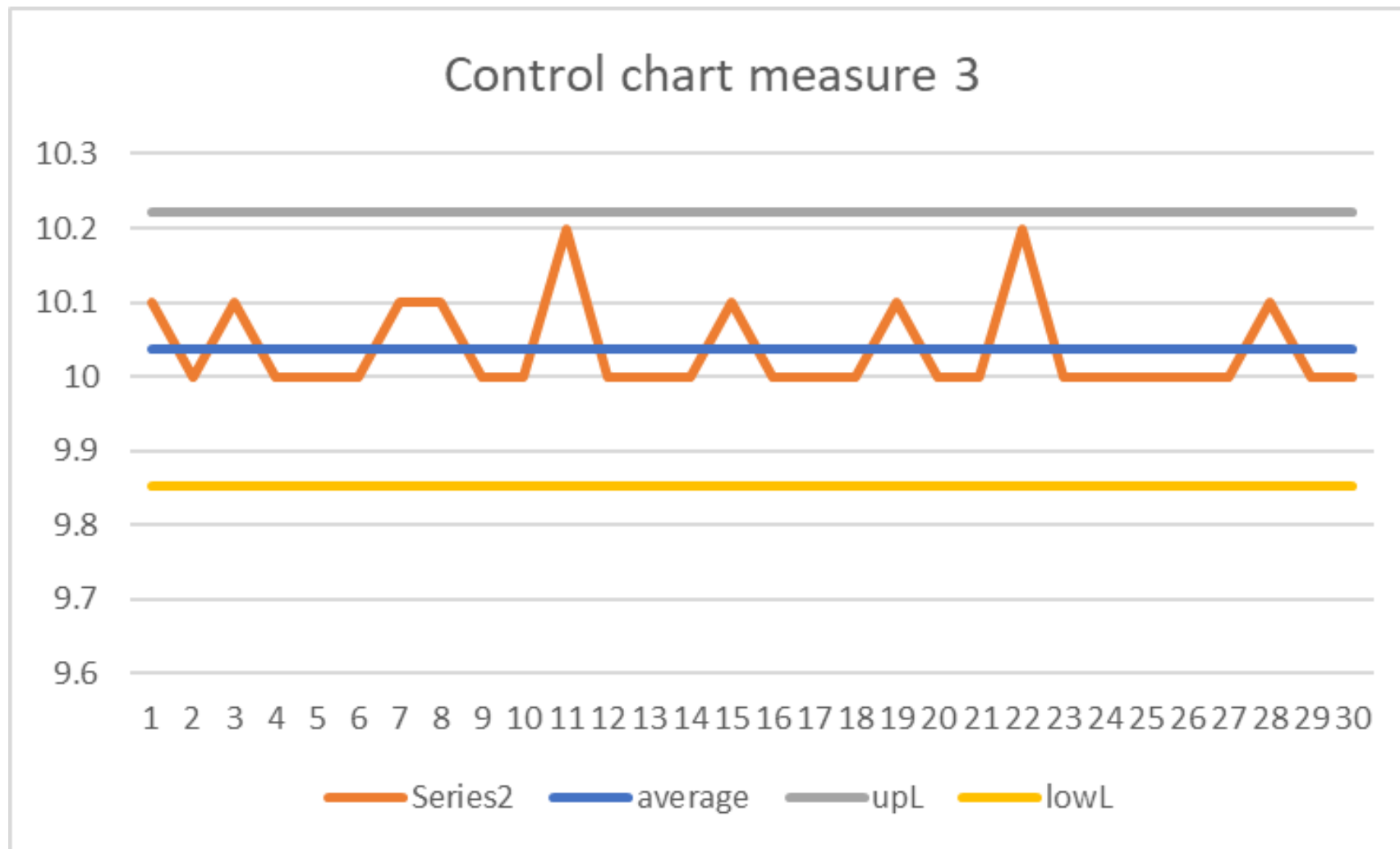
a)



b)

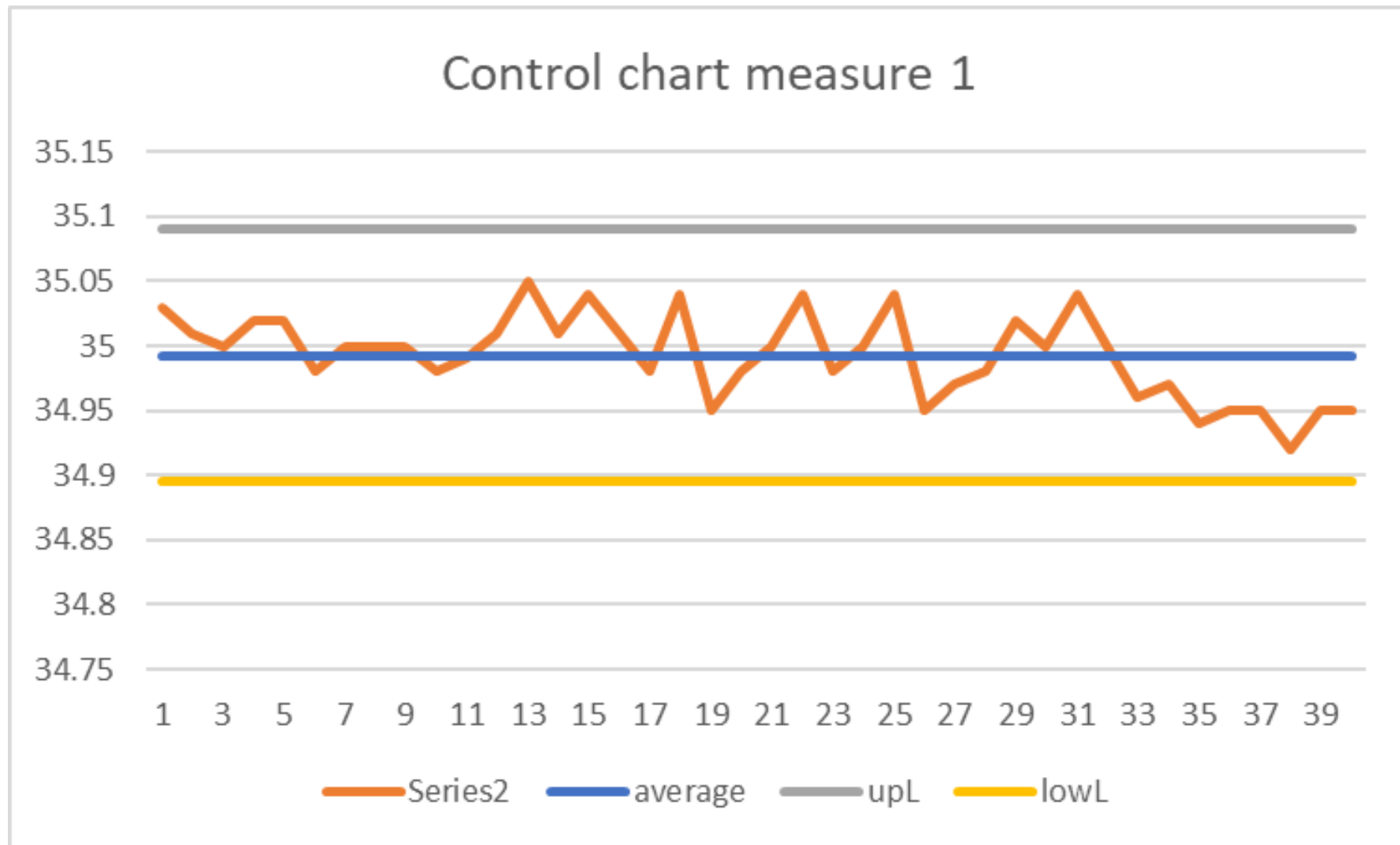


c)

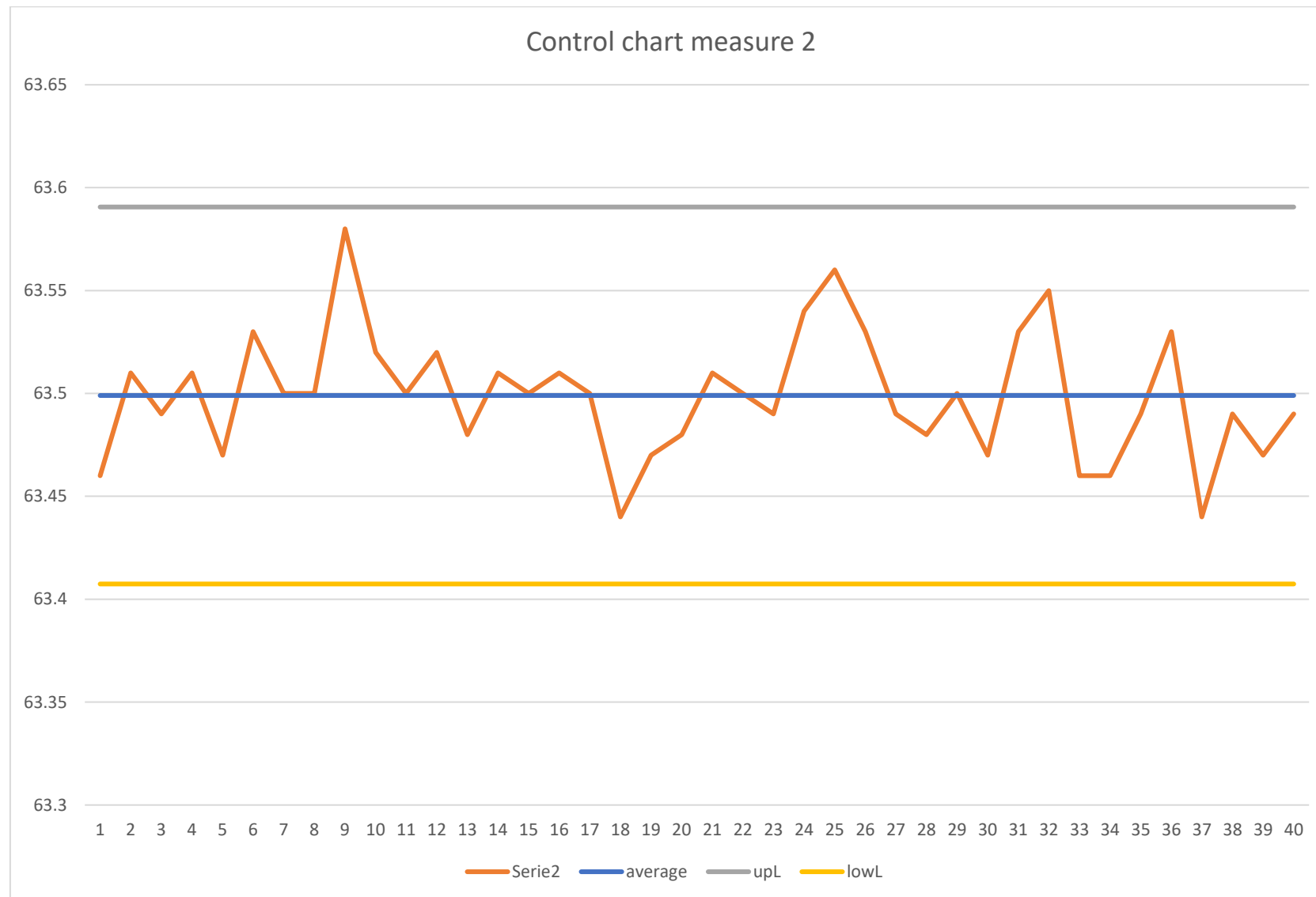


URC 176

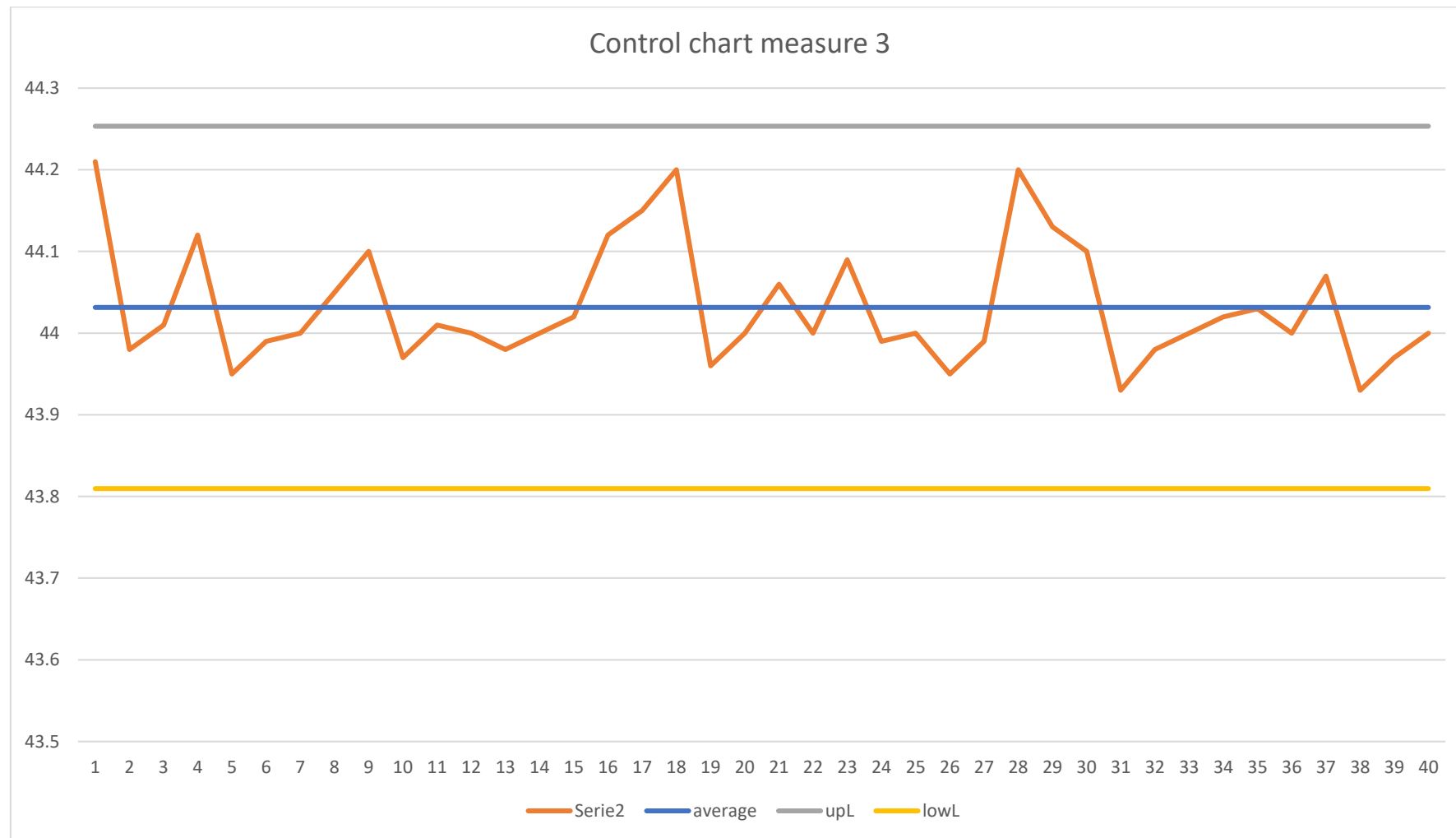
a)



b)



c)



Conclusions:

According to the control charts we can see that the measured pieces' lengths are all within the UpL and LowL limits, that means we are in control.

In Conclusion, the incorporation of control charts in Del Morino's factory plays a vital role in achieving superior production standards, streamlining manufacturing processes, and bolstering the company's reputation as a trustworthy supplier of high-quality farm equipment worldwide. The continuous analysis of process performance and data-driven improvements enable Del Morino to remain at the forefront of innovation and effectively meet the dynamic demands of the agricultural industry

References:

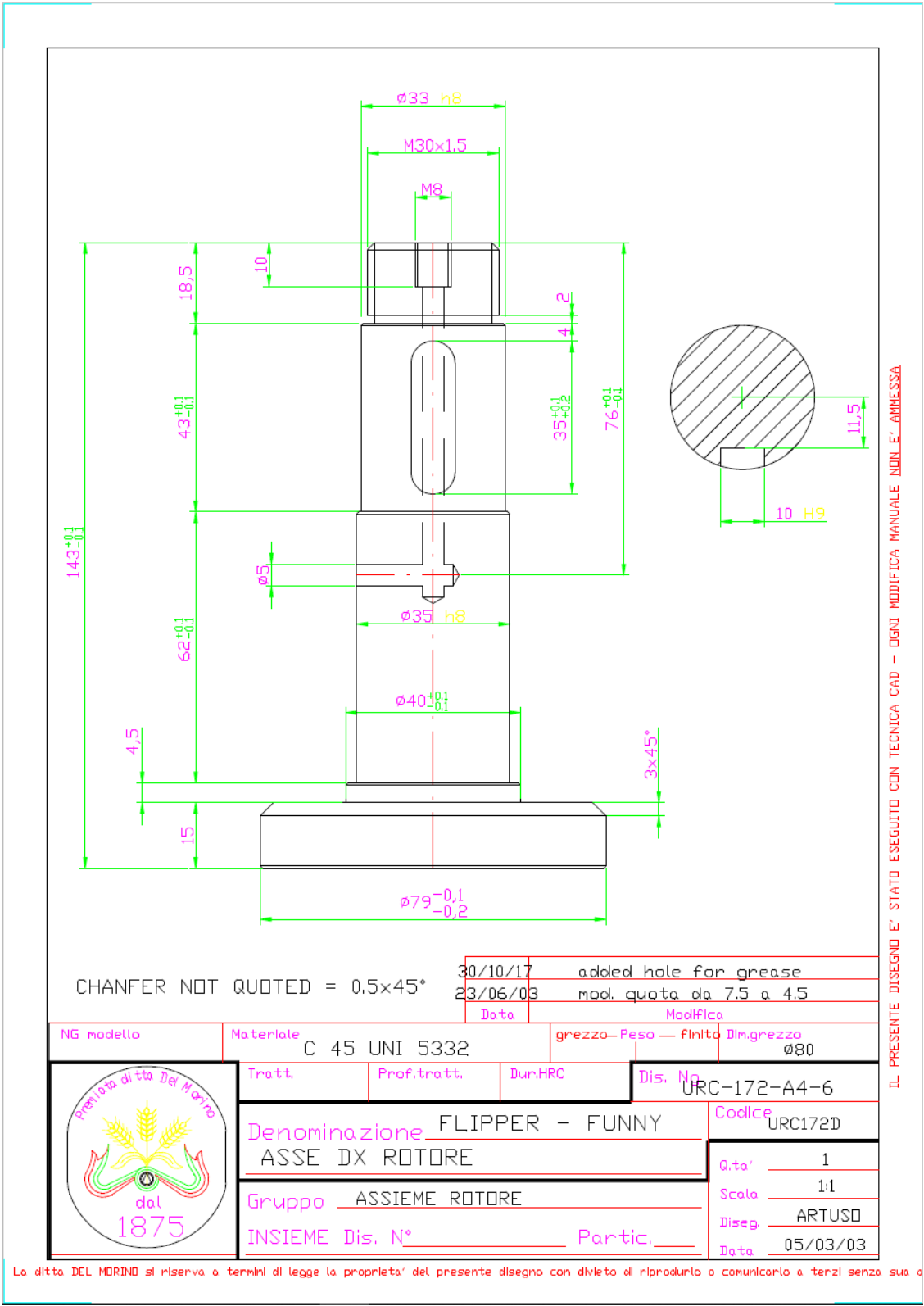
https://digfir-published.macmillanusa.com/psbe4e/psbe4e_ch12_5.html

<https://www.simplilearn.com/what-is-quality-control-article>

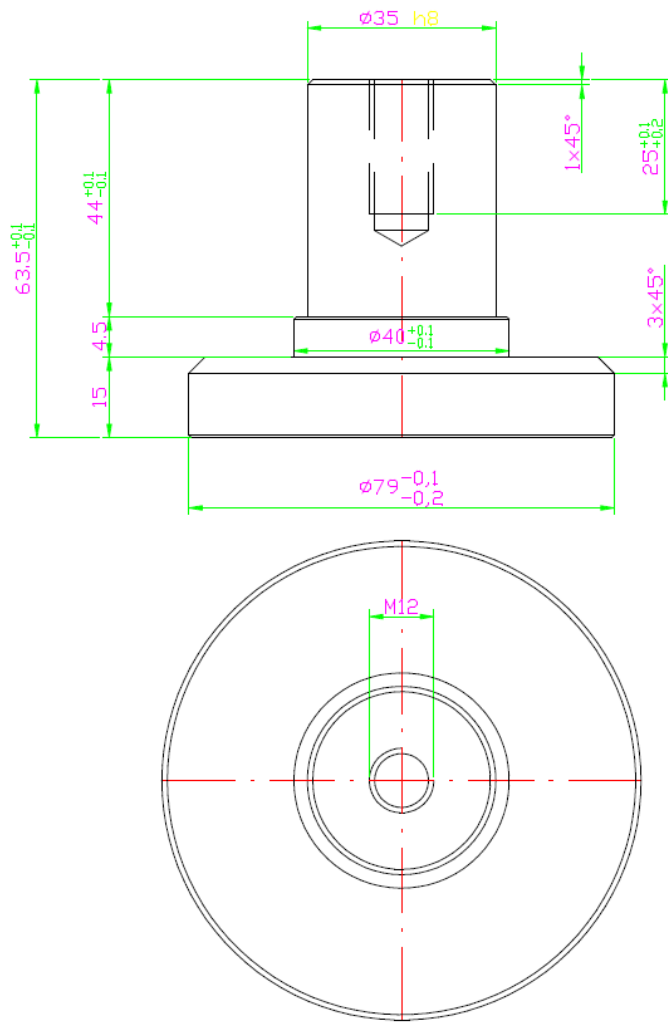
<https://asq.org/quality-resources/control-chart>

Annex A:

URC172



URC176



CHANFER NOT QUOTED = 0.5x45°

23/06/03		mod. quota da 7.5 a 4.5	
Data		Modifica	
NG modello	Materiale	grezzo-Peso	Dim.grezzo
	C 40 UNI 5332	— finito	Ø80
	Trett.	Prof.trett.	Dur.HRC
	Dis. No		URC-176-A4-4
	Denominazione		Code
	FLIPPER - FUNNY		URC176D
	ASSE SX ROTORE		
Gruppo		Q.ta'	1
ASSIEME ROTORE		Scala	1:1
INSIEME Dis. N°		Diseg.	ARTUSO
URC-056-A0-0		Data	05/03/03
Partic.			

IL PRESENTE DISEGNO E' STATO ESEGUITO CON TECNICA CAD - OGNI MODIFICA MANUALE NON E' AMMESSA

La ditta DEL MORINO si riserva a termini di legge la proprietà del presente disegno con divieto di riprodurlo o comunicarlo a terzi senza sua autorizzazione.